Database Management System

Assignment Report

Group 2: Library Database Management System (LMS)

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Introduction

This report provides a comprehensive summary of all components involved in Task 1. It outlines each element clearly, offering supporting details and justifications to demonstrate how they contribute to the overall task. The content is not just designed to meet the task requirements but also to help students understand how a database system works.

The report covers an overview of the selected industry, a description of the database system used, the design of Entity-Relationship Diagram (ERD) and the database implementation. It also provides the real-life example of the database system used in the sector, and shows how the system helps these companies manage their operations efficiently. In addition, the report discusses current limitations of the database system and proposes practical suggestions for improvement to enhance its data handling, performance, scalability and integration .

Overall, the report ensures a strong understanding of how the database system works and how it can be used in real life. Through detailed analysis, structured design and practical application, the report shows how the database system can improve the management and organise information. Everything in the report is designed to meet the project goals and criteria effectively.

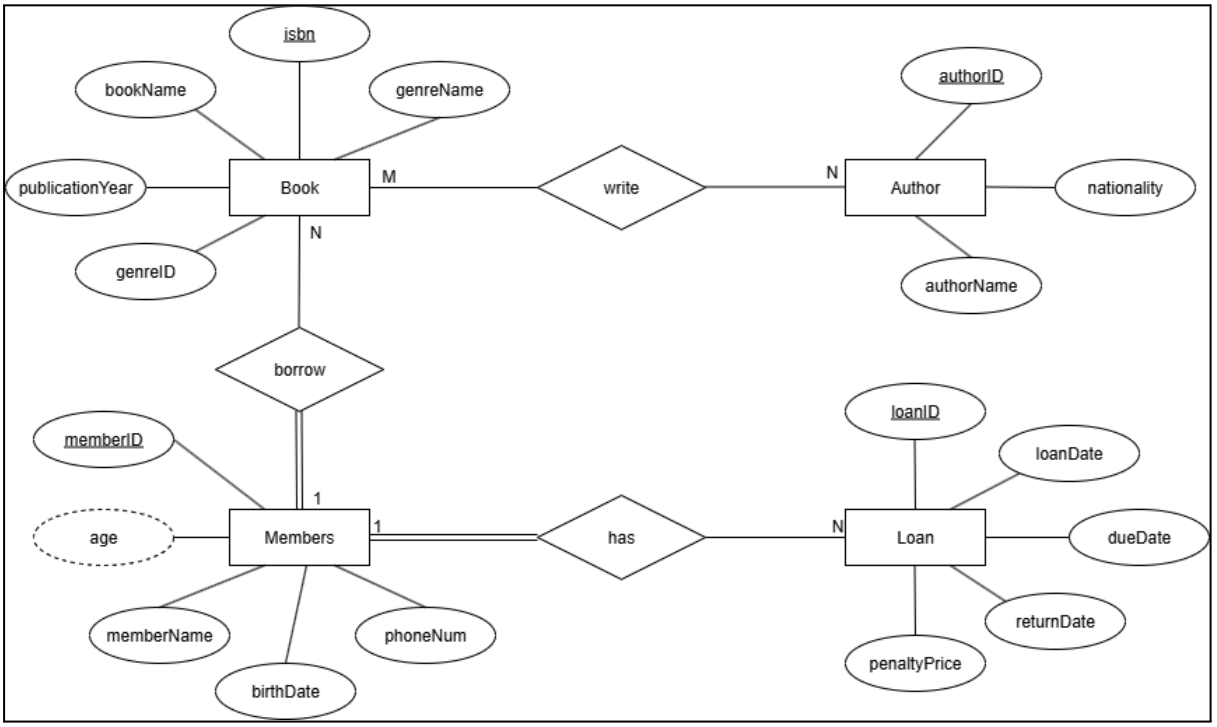
Task 1.1 System Design

System: Library Management System (LMS)

Identification of the Database System or Process
A Library Management System is a system to manage the library and track books available by looking at the ISBN, Author Name, Title, etc. This also includes a system to track members if they are loaning any books, the loan date and due date and or any penalties charged.

How it supports the selected industry
LMS helps libraries to run better such as ensuring data accuracy, if the book's information is wrong, instead of finding it manually we can search it by using CRUD operations to modify the information.

Entity Relationship Diagram (ERD)



Created using draw.io

The following are the relationships between the entities:

- An author can write many books and a book can belong to many authors.
- A member can borrow many books, but each book must only belong to one member.
- A member can have many loans, but a loan must only belong to one member.

Components

Book:
Attributes: isbn (PK), bookName, publicationYear, genreName, genreID
Purpose: Store the book's information.

Members:
Attributes: memberID (PK), age (derived attribute), memberName, phoneNum, birthDate
Purpose: Store the member's information

Author:
Attributes: authorID (PK), authorName, nationality
Purpose: Store the author's details

Loan:
Attributes: loanID (PK), loanDate, dueDate, returnDate, penaltyPrice
Purpose: Keep track of the member's loaning information

A brief explanation:

The Members entity is for the visitor who wants to apply for a membership so they can loan a book. The loan entity is to track which book and when the book is loaned and whether there is any penalty given. The Author entity is information about the respective authors and lastly the Book entity is information of the books.

Task 1.2 Database Implementation

Based on the Chen notation ERD in Task 1.1 System Design, the diagram has not gone through normalization. Normalization is the process of organizing data in a database to reduce redundancy and improve data integrity.

This process is important while coming up with a new database system as it will bring significant benefits such as reducing data redundancy, saves storage space, help create well-structured tables and more.

1NF - to remove atomic values and multivalue attributes
In this case, there are no multivalue attributes

Book	
PK	isbn
	bookName
	publicationYear
	genreID
	genreName

Loan	
PK	loanID
	loanDate
	dueDate
	returnDate
	penaltyPrice

Members	
PK	memberID
	memberName
	phoneNum
	birthDate
	age

Author	
PK	authorID
	authorName
	nationality

Created using Microsoft Excel

1NF passed ✓

2NF - to remove partial dependencies

In the book table, genreName only relies on genreID, hence it is a partial dependency

A new junction table BookAuthor is also created due to the table Book and Author having many-to-many relationship

3NF

Book	
PK	isbn
	bookName
	publicationYear
FK	genreID
FK	memberID

Genre	
PK	genreID
	genreName

Members	
PK	memberID
	memberName
	phoneNum
	birthDate
	age

Loan	
PK	loanID
	loanDate
	dueDate
	returnDate
	penaltyPrice
FK	memberID

Author	
PK	authorID
	authorName
	nationality

BookAuthor	
PK, FK	isbn
PK, FK	authorID

Created using Microsoft Excel

2NF passed ✓

3NF - to remove transitive dependencies

There are no transitive dependencies in this case as there aren't any non-key attributes that rely on any non-key attributes

Hence, no changes

3NF

Book	
PK	isbn
	bookName
	publicationYear
FK	genreID
FK	memberID

Genre	
PK	genreID
	genreName

Members	
PK	memberID
	memberName
	phoneNum
	birthDate
	age

Loan	
PK	loanID
	loanDate
	dueDate
	returnDate
	penaltyPrice
FK	memberID

Author	
PK	authorID
	authorName
	nationality

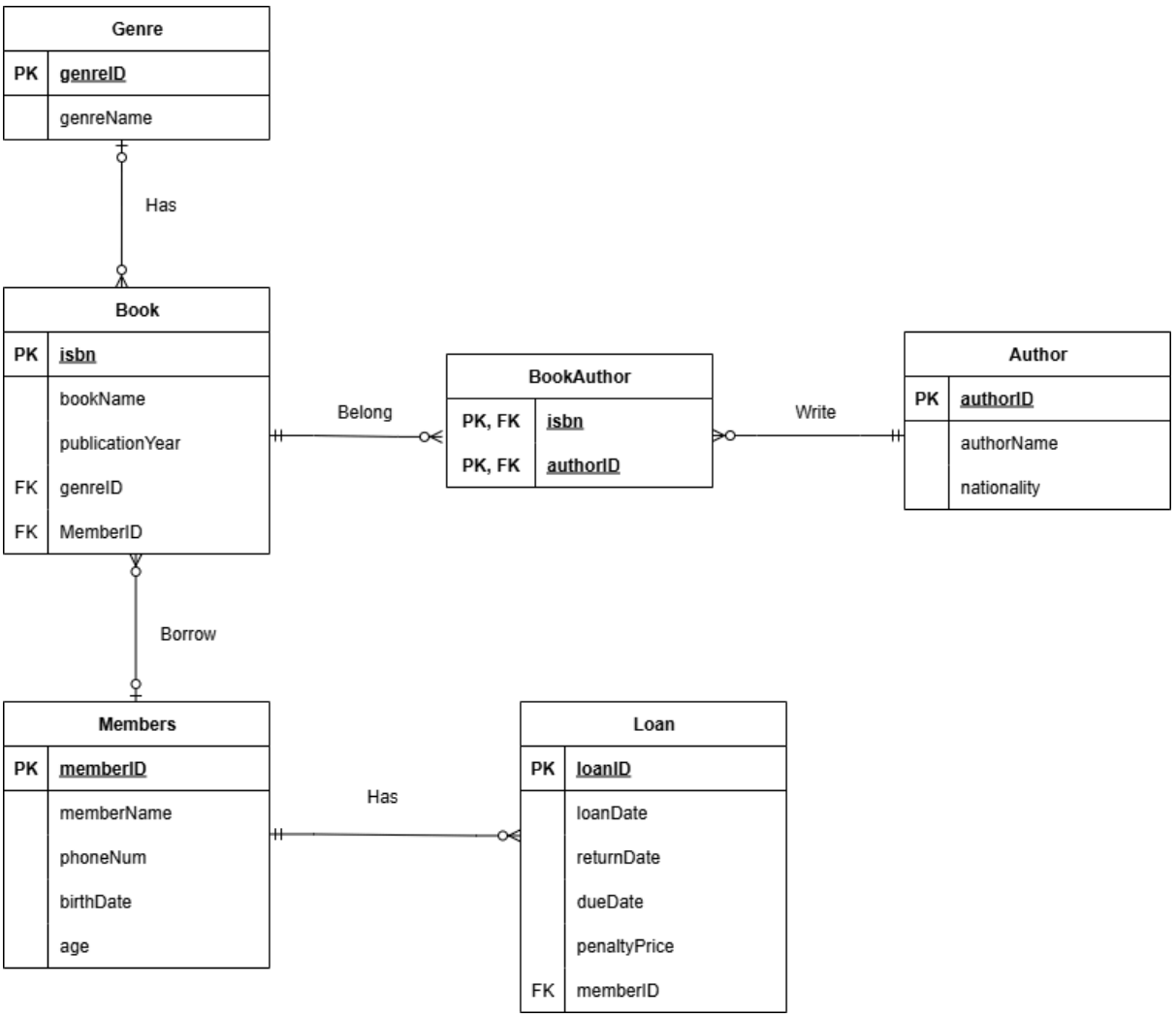
BookAuthor	
PK, FK	isbn
PK, FK	authorID

Created using Microsoft Excel

3NF passed ✓

Crow's Foot Diagram

According to the 3NF's result, a crow's foot diagram is created



Created using draw.io

SQL Commands Used

A) CRUD Operations

```
-- inserting data into tables (CREATE)
INSERT INTO members (memberID, memberName, phoneNum, age)
VALUES
('CR001', 'Bobby Meng Leng', '019-2321123', 20),
('CR002', 'Jennifer Lee', '016-9001033', 18);

INSERT INTO author (authorID, authorName, nationality)
VALUES
('RS0001', 'J.K. Rowling', 'United Kingdom'),
('RS0002', 'Stephen King', 'United States'),
('RS0003', 'Victor Axelson', 'Denmark');
```

```
-- Viewing the data (READ)
SELECT * FROM members
WHERE age = 20;

SELECT * FROM author;
```

```sql
-- Updating the data (UPDATE)
UPDATE members
SET age = 22
WHERE memberID = 'CR002';

UPDATE author
SET nationality = 'France', authorName = 'Jean Paul Gaultier'
WHERE authorID = 'RS0001';
```

```sql
-- Deleting data (DELETE)
DELETE FROM author
WHERE authorID = 'RS0003';

DELETE FROM members
WHERE memberID = 'CR002' ;
```

B) Advanced SQL Commands

```sql
-- Views (Jordan)
CREATE VIEW book_genre_view AS
SELECT book.isbn, book.bookName, genre.genreName
FROM book
JOIN genre ON book.genreID = genre.genreID;

SELECT * FROM book_genre_view;
```

Hide unnecessary columns and provide easier access to related data.

```sql
-- Function (Zahran)
DELIMITER \\
CREATE FUNCTION GetMembersAge(birthDate DATE)
RETURNS INT
DETERMINISTIC
BEGIN
    RETURN TIMESTAMPDIFF(YEAR, birthDate, CURDATE());
END\\

DELIMITER ;
```

A function to calculate the derived attribute (age) from birthDate

```sql
-- Stored procedure (Lok Thim)

DELIMITER \\

CREATE PROCEDURE addAuthor (
    IN p_authorID VARCHAR(50),
    IN p_authorName VARCHAR(100),
    IN p_nationality VARCHAR (50)
)
BEGIN
    INSERT INTO author (authorID, authorName, nationality)
    VALUES (p_authorID, p_authorName, p_nationality);
END\\

DELIMITER ;

CALL addAuthor ('RS004', 'Stan Lee', 'United States');
```

This stored procedure is created to enable easier and faster data entry for the author table

```sql
-- Trigger (Edison)

DELIMITER \\
CREATE TRIGGER after_member_insert
AFTER INSERT ON members
FOR EACH ROW
BEGIN
  INSERT INTO member_log (memberID, memberName, action_type)
  VALUES (NEW.memberID, NEW.memberName, 'INSERT');
END\\

DELIMITER ;
```

The trigger automatically tracks new member additions by recording their ID and name into a member_log table right after they're inserted into the members table.

```
-- Trigger (George)

DELIMITER \\
CREATE TRIGGER set_penalty_before_update
BEFORE UPDATE ON loan
FOR EACH ROW
BEGIN
    IF NEW.returnDate > NEW.dueDate THEN
        SET NEW.penaltyPrice = DATEDIFF(NEW.returnDate, NEW.dueDate) * 1.50;
    END IF;
END\\
DELIMITER ;
```

This trigger is activated **before** a loan record is updated. If the return date is **later** than the due date, it calculates the number of late days and multiplies it by RM1.50 to set the "penaltyPrice".

C) Restricting and Sorting SQL



authorID	authorName	nationality
RS0002	Stephen King	United States
RS004	Stan Lee	United States
NULL	NULL	NULL

SELECT * FROM author WHERE authorName LIKE 'S%';



authorID	authorName	nationality
RS0002	Stephen King	United States
RS004	Stan Lee	United States
RS0001	Jean Paul Gaultier	France
NULL	NULL	NULL

SELECT * FROM author ORDER BY nationality DESC;

Single Group Function in SQL

	memberName	upper_memberName
▶	Bobby Meng Leng	BOBBY MENG LENG

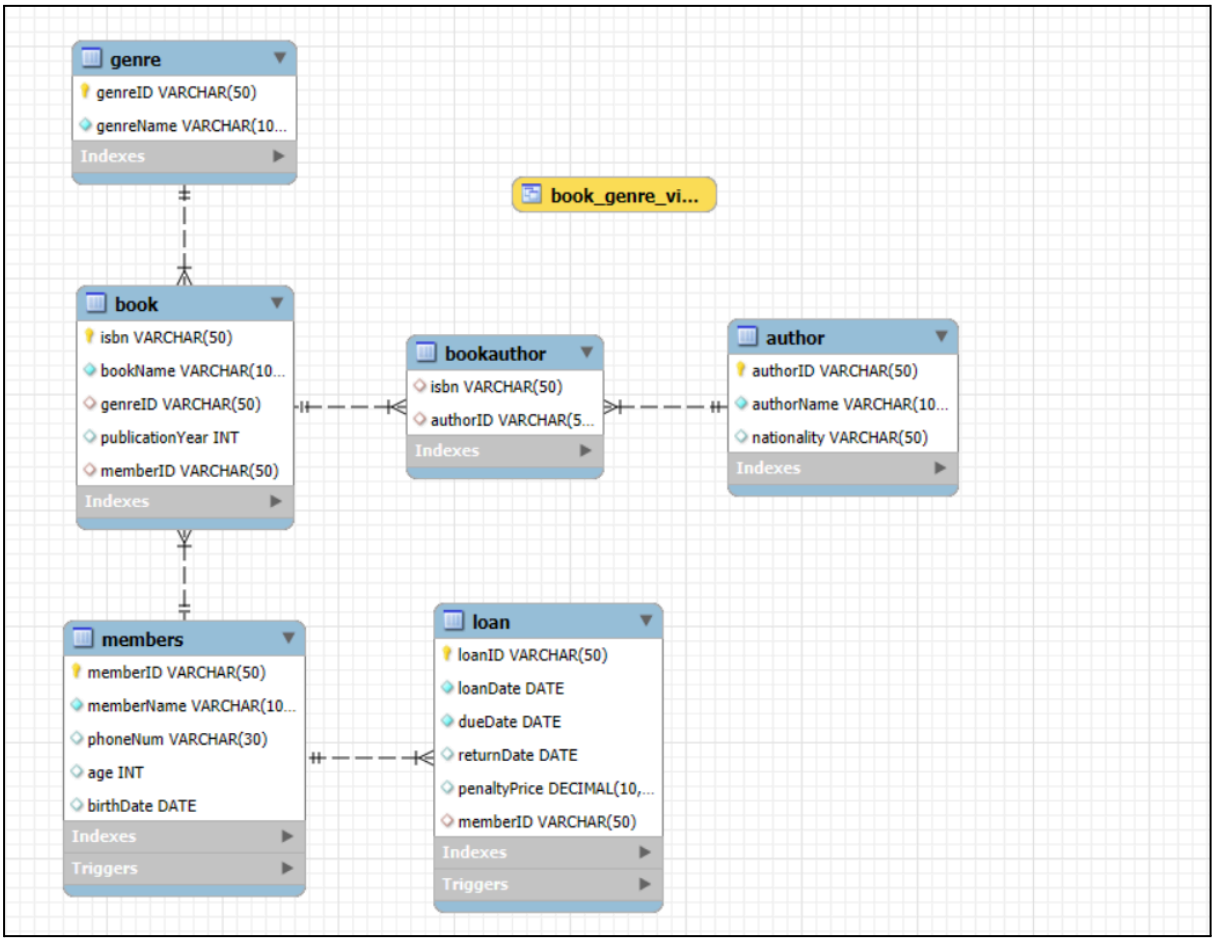
SELECT memberName, UPPER(memberName) AS upper_memberName FROM members;

Result Grid | Filter Rows:

	authorName	author_shortName
▶	Jean Paul Gaultier	Jean Pa
	Stephen King	Stephen
	Stan Lee	Stan Le

SELECT authorName, SUBSTRING(authorName, 1, 7) AS author_shortName FROM author;

MySQL's Crow Foot Diagram based on Reverse Engineer



The diagram above shows how MySQL has created a crow foot diagram based on our current database structure using Reverse Engineer.

Task 1.3 Real Life Examples

A Library Management System (LMS)

A Library Management System (LMS) is a software program that automates all the operations within a library including acquisition, cataloguing, circulation, access control, and reporting. LMS is used across corporations, research institutes, hospitals and public institutions to manage both digital and physical resources. A nice example is the Library of University Malaya.

Universiti Malaya Library, one of the most prestigious academic libraries in Malaysia, uses an integrated LMS system called Virtua ILS, which is built on a relational DBMS from a multi-million corporation. It tracks various academic resources. This system is responsible for:

- Automating the borrowing and returning of items from the library
- Maintaining real-time records of inventory and records of user borrowing activities
- Reporting user activities and trends including overdue books

The system ensures data consistency, fast query response time, and role-based access control for students, librarians, and administrators. DBMS is particularly important for efficient library operations; centralizing data provides easy access for multiple branches and users. This automates repetitive tasks like calculating fines, generating overdue notices, and managing reservation queues, greatly lowering human error and saving operating time (GeeksforGeeks, 2024).

Furthermore, a DBMS offers robust analytics, allowing libraries to make calculated decisions on acquisitions, staffing, and resource allocation. Advanced SQL techniques—such as stored procedures, triggers, and views—play a key role in supporting these features and ensuring operational efficiency (Mudgal, 2023). In practical LMS implementations, triggers are often used to automatically calculate penalties for overdue books and update inventory counts without manual intervention (GeeksforGeeks, 2023).

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Task 1.4 Limitations And Suggestions For Improvement

Part 1 - University of Malaya (Virtua ILS)

Task 4(part1) Enhancement/Innovations that could improve data handling,scalability, performance or integration.

Universiti Malaya Library uses Virtua ILS, a reliable integrated system based on a relational DBMS. It supports key functions such as automated borrowing and returns, real-time inventory tracking, user activity reporting, and access through student portals. However, despite its effectiveness, there are some limitations and opportunities for enhancement:

Current limitations

1. Scalability issues

   Relational database management systems may encounter difficulties managing increasing data volumes as the number of digital resources and users rises.

2. Limited Ability to Integrate

   Accessibility and user convenience may be diminished by the system's difficulty connecting to contemporary resources like external journal databases, mobile applications, and e-learning platforms.

3. High Maintenance Costs

   The library management system will be more expensive to operate and less flexible in the event of future changes because the vendor is in charge of system updates, maintenance, and customizations.

Proposed enhancements/ Innovations

1. Migration to Cloud database

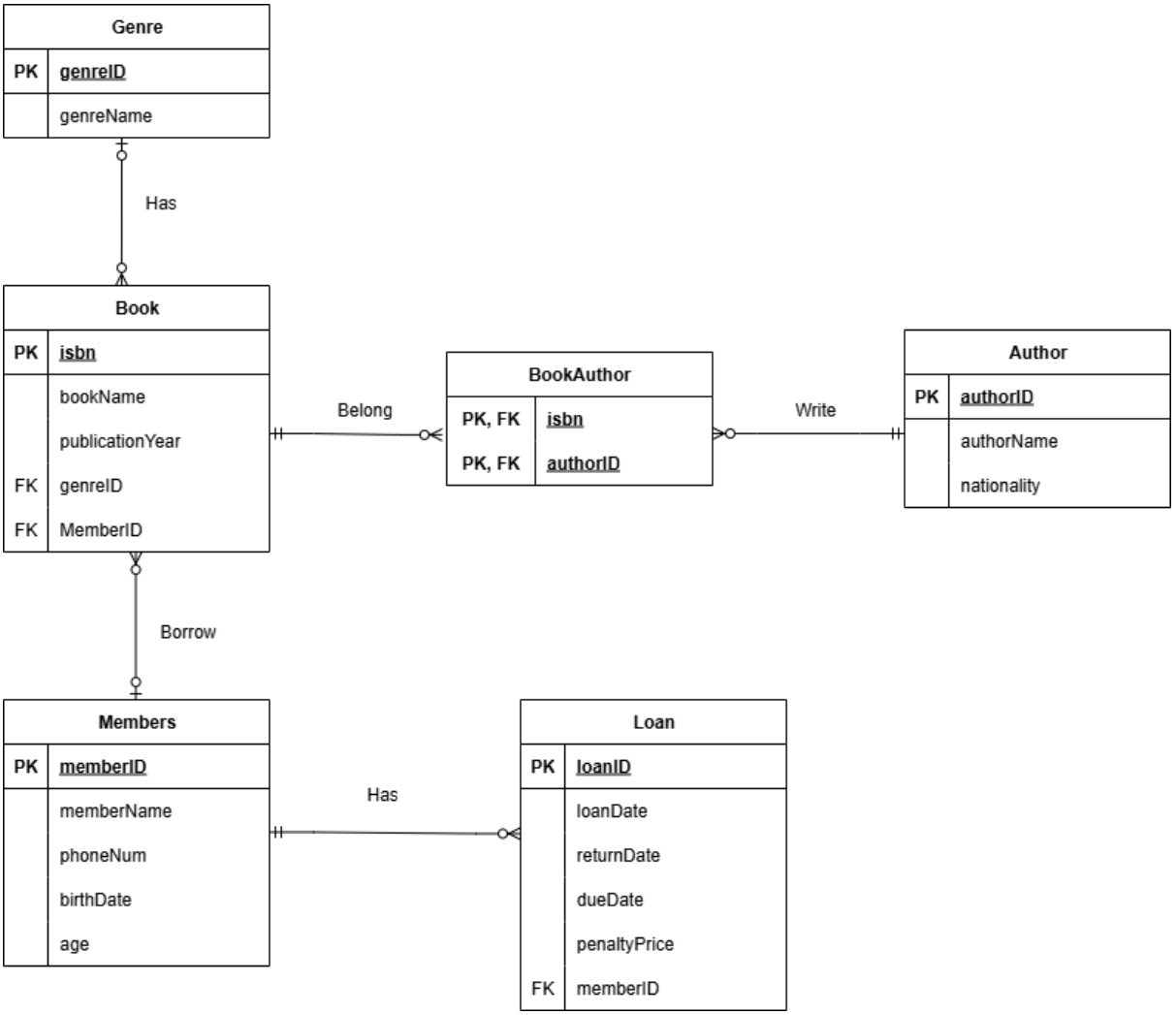
   Moving the library's database over to a cloud-based system like Google Cloud would make it easier for the staff to manage, decrease maintenance costs, scalability and improve its performance.

2. Implementation of APIs

   Creating APIs would make it possible to integrate other platforms like mobile apps, learning management systems, and third-party digital libraries more easily.

Part 2 - Current Assignment's LMS

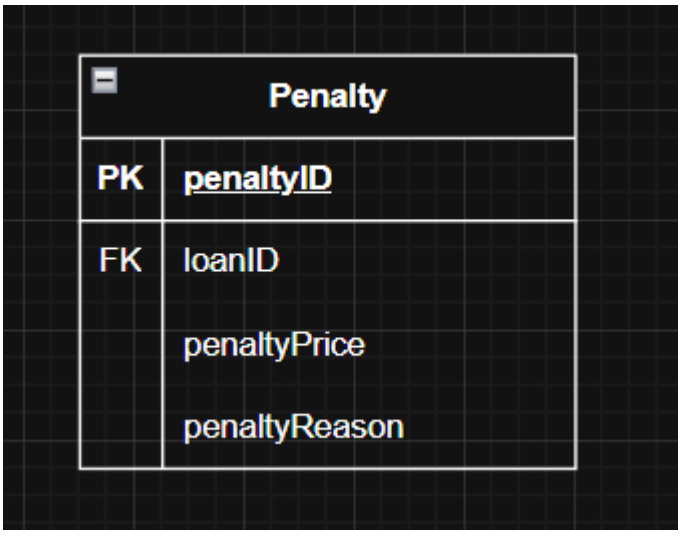
Judging the current crow's foot diagram, there are a few improvements able to be added to our system to ensure a more detailed and smooth process for users. It is always advisable to find weaknesses from our work to understand our weak points and overcome the issues.



1. Adding a Penalty table

   When we add a new Penalty table, we can also include more details for the penalties, such as the amount and reason of the penalty.

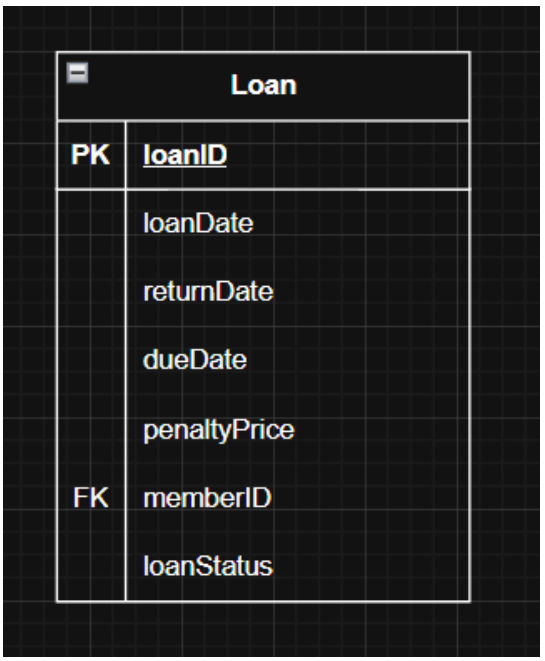
   Example:



2. Adding a status attribute for the Loan table

   The addition should give the status of the book such as loaned, returned and overdue. This can help users to know more about the book.

   Example:



This is the end of the report. Thank you for reading :)



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